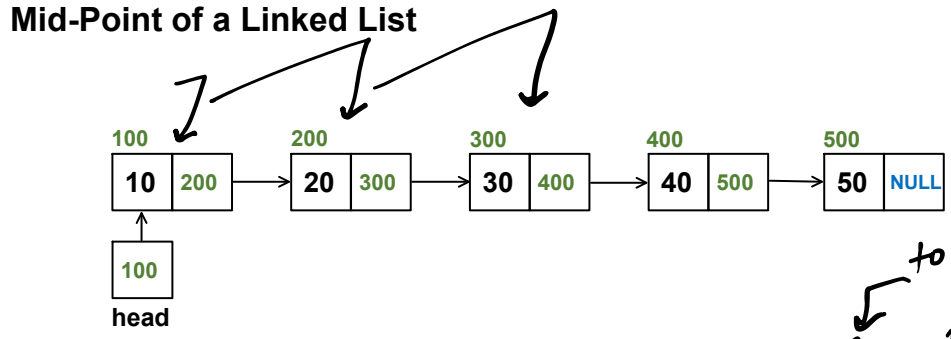
$5/2 = 2$

$n = 5$

Mid-Point of a Linked List



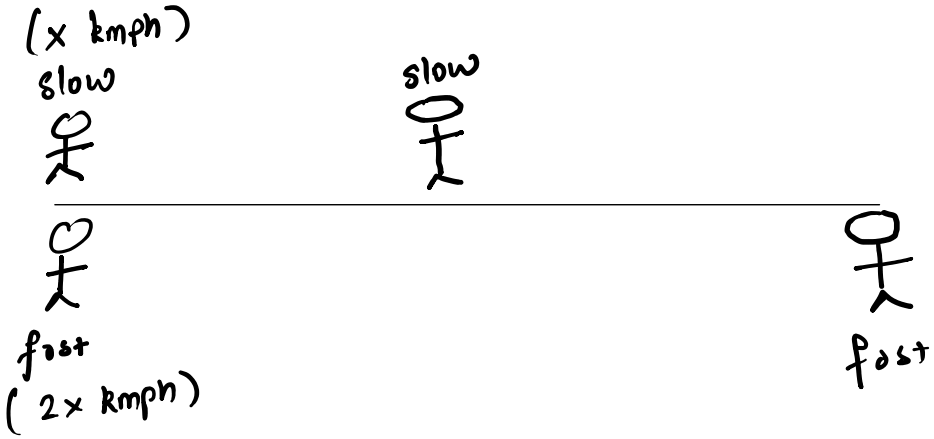
to find the len of the LL
to get the mid-point

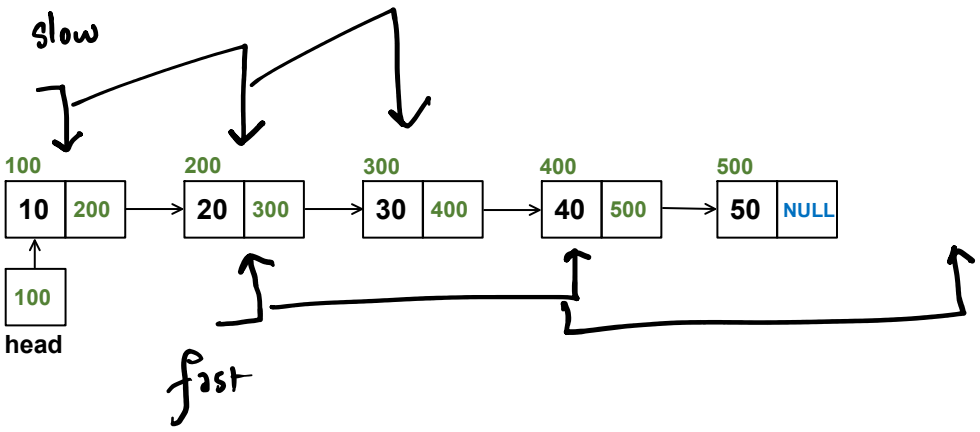
$n + \frac{n}{2} = \frac{3n}{2}$. const

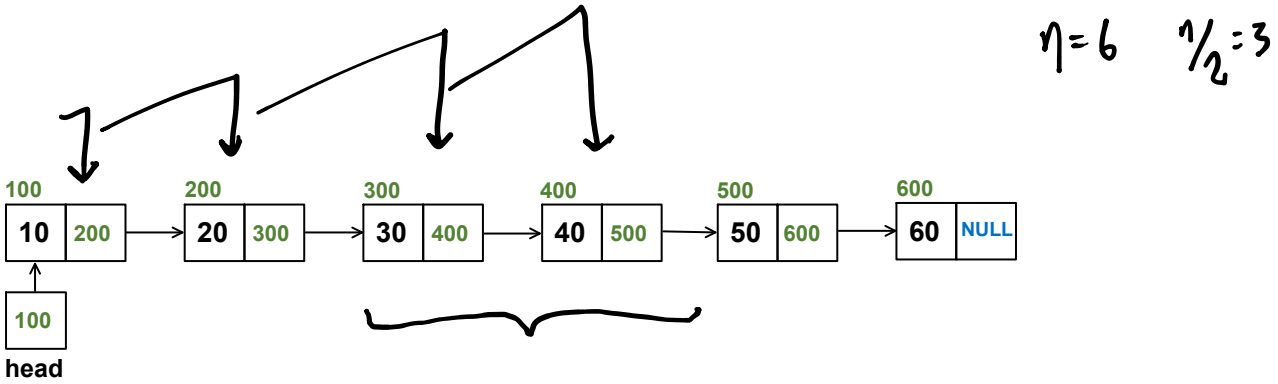
tortoise - hare
(slow) (fast)

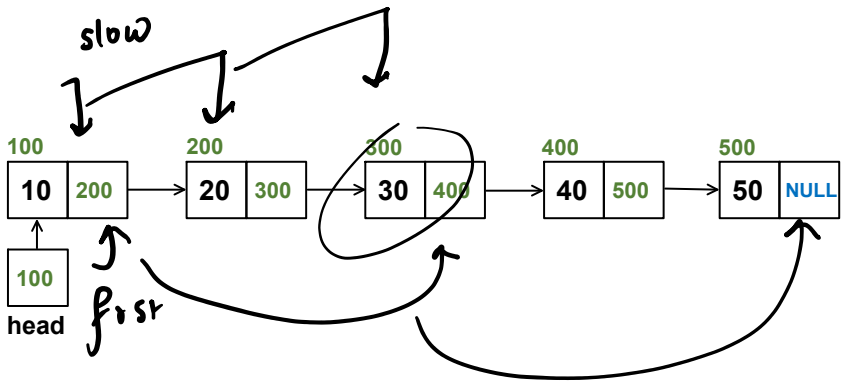
time: $O(n)$

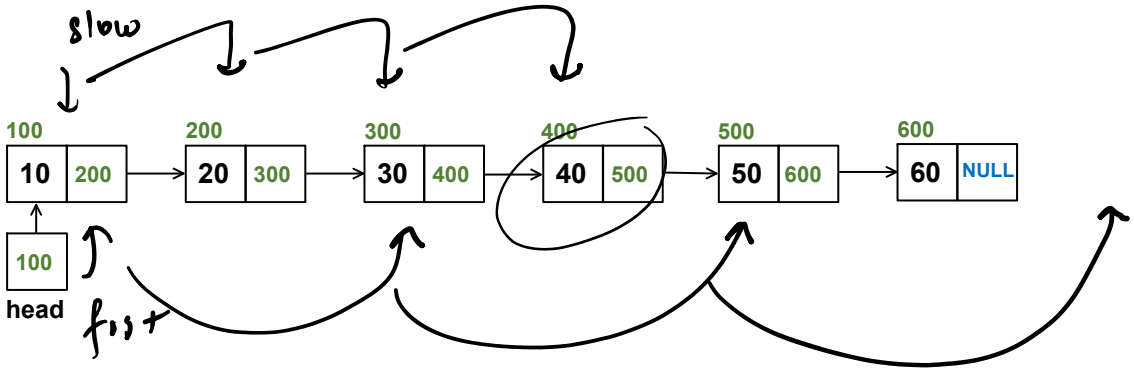
Slow-fast ptr appr






n=6 n/2=3
100
10
200
20
300
30
400
40
500
50
600
60
NULL
100
head


slow
100
10
200
20
300
30
400
40
500
50
NULL
100
head
fast

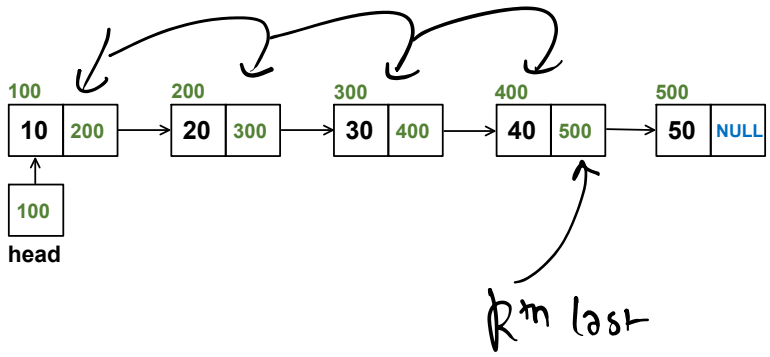

slow
100
10
200
20
300
30
400
40
500
50
600
60
NULL
100
head
fast

n = 5

50 → 40 → 30 → 20 → 10

K = 2

K^th Node from End in a Linked List



K^th last node

appr 1

1. find the length of the LL = n
2. Move n-K steps from head = n-K

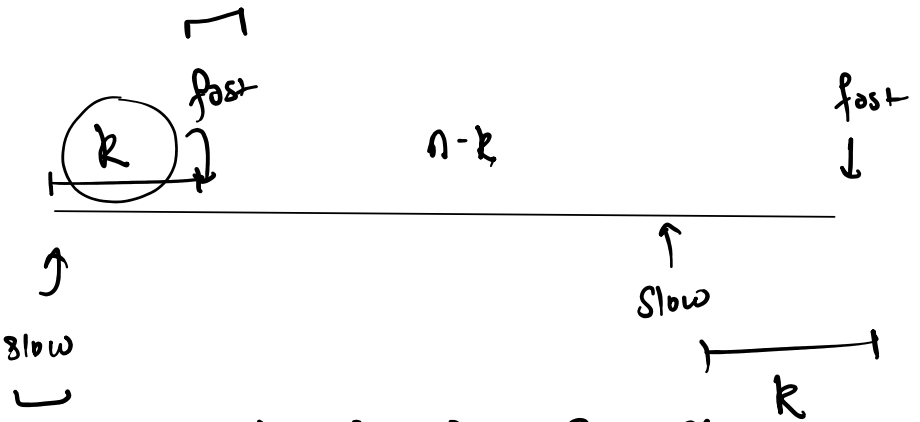
2n-K

appr 2

1. Reverse the LL = n
2. Move k-1 steps from head = k-1

n + k - 1

appr. 3



k + (n-k) = n steps

move fast by K-steps

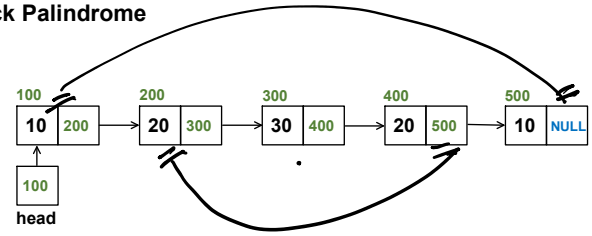
move slow & fast

move 1st
by k-steps
and initialise
slow with
head

move slow & fast
at the same speed
till fast becomes
null, at this point
slow will be at the
kth last node

Check Palindrome

i/p :



(HW)
1. create a clone of given LL => n
2. reverse the clone => n
3. cmp with org. ll => n

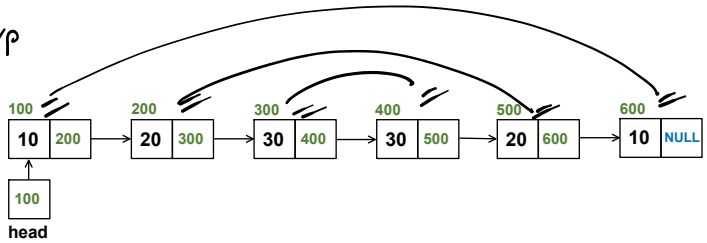
3n steps

time : O(n)

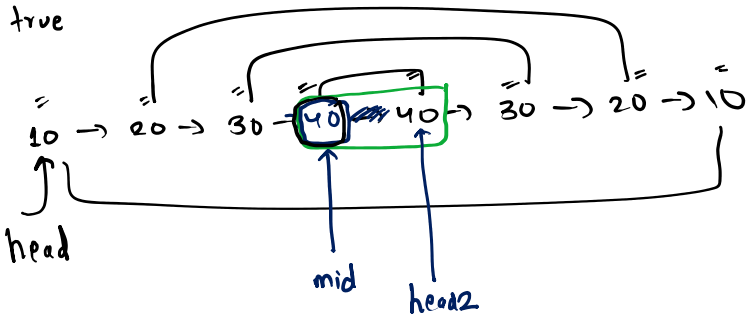
space : O(n) due to clone

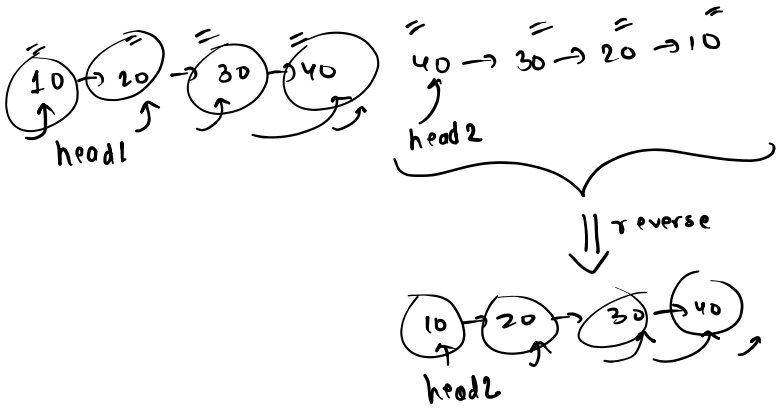
o/p : true

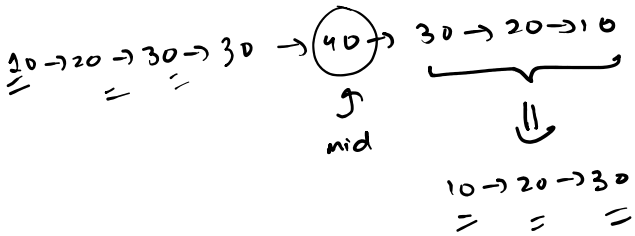
i/p



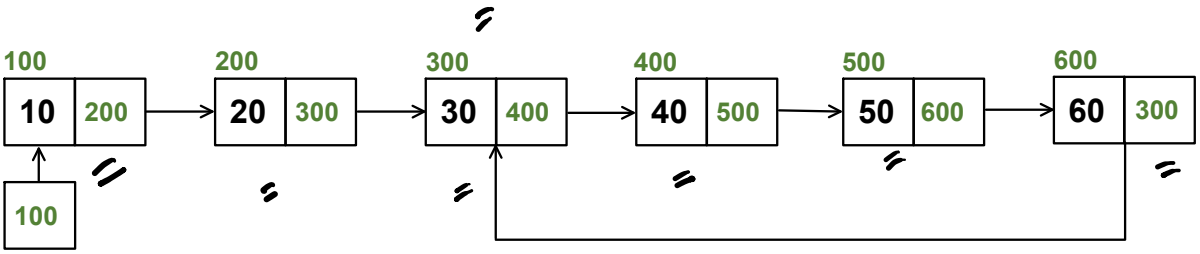
o/p : true

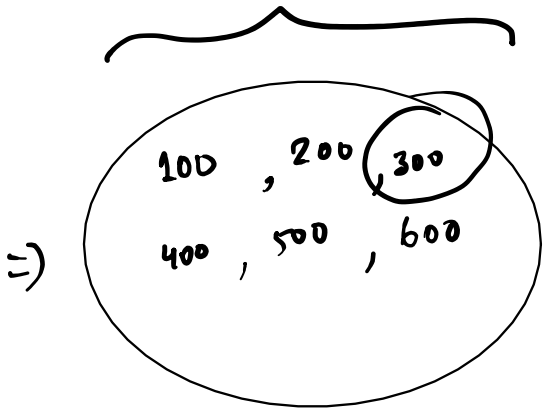


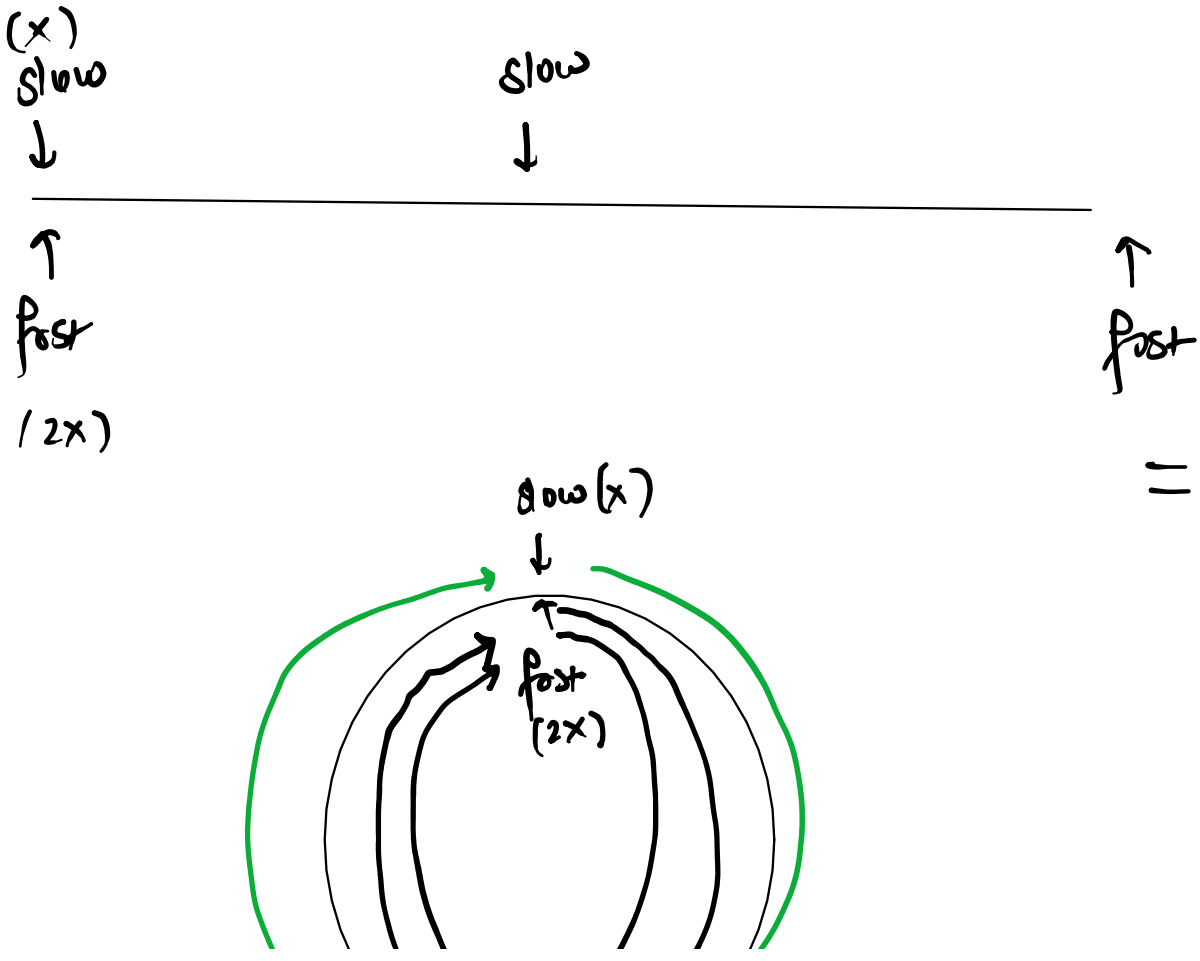


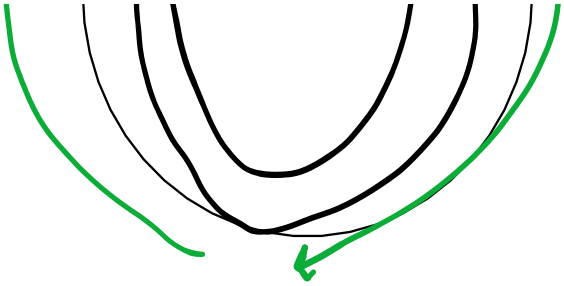


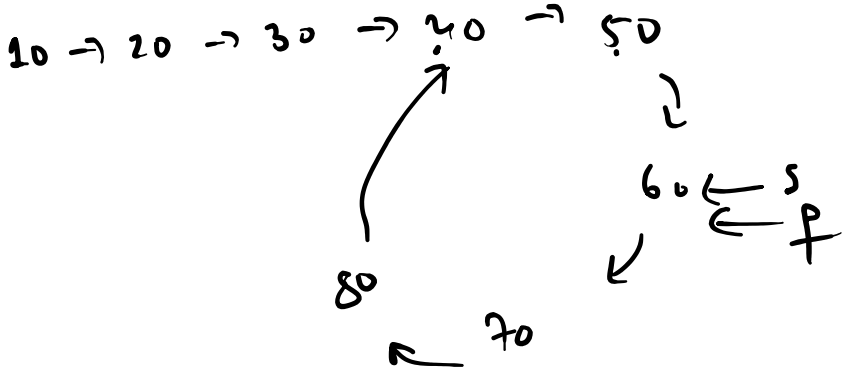
Detect Cycle in a Linked List

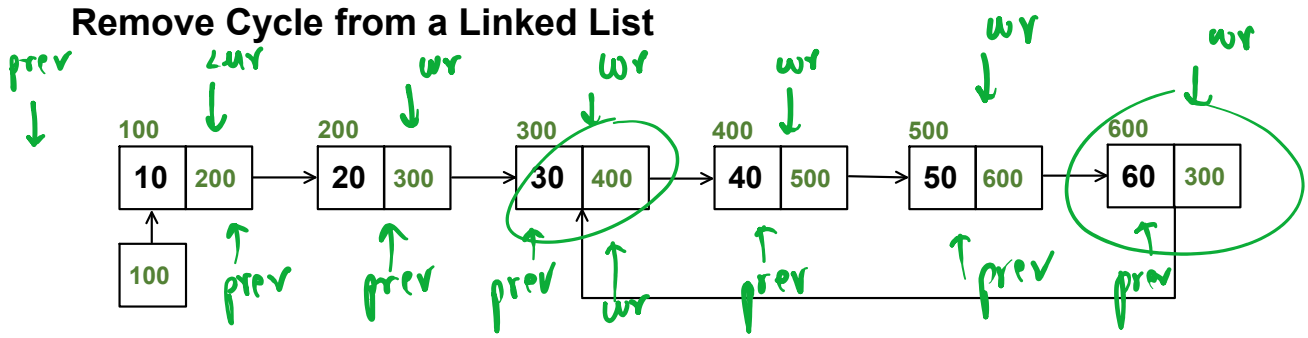


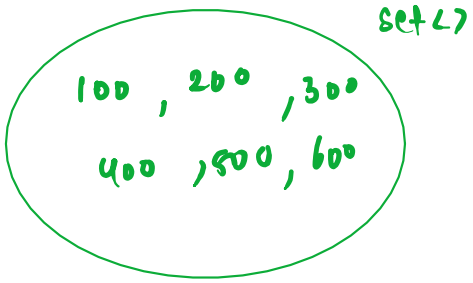


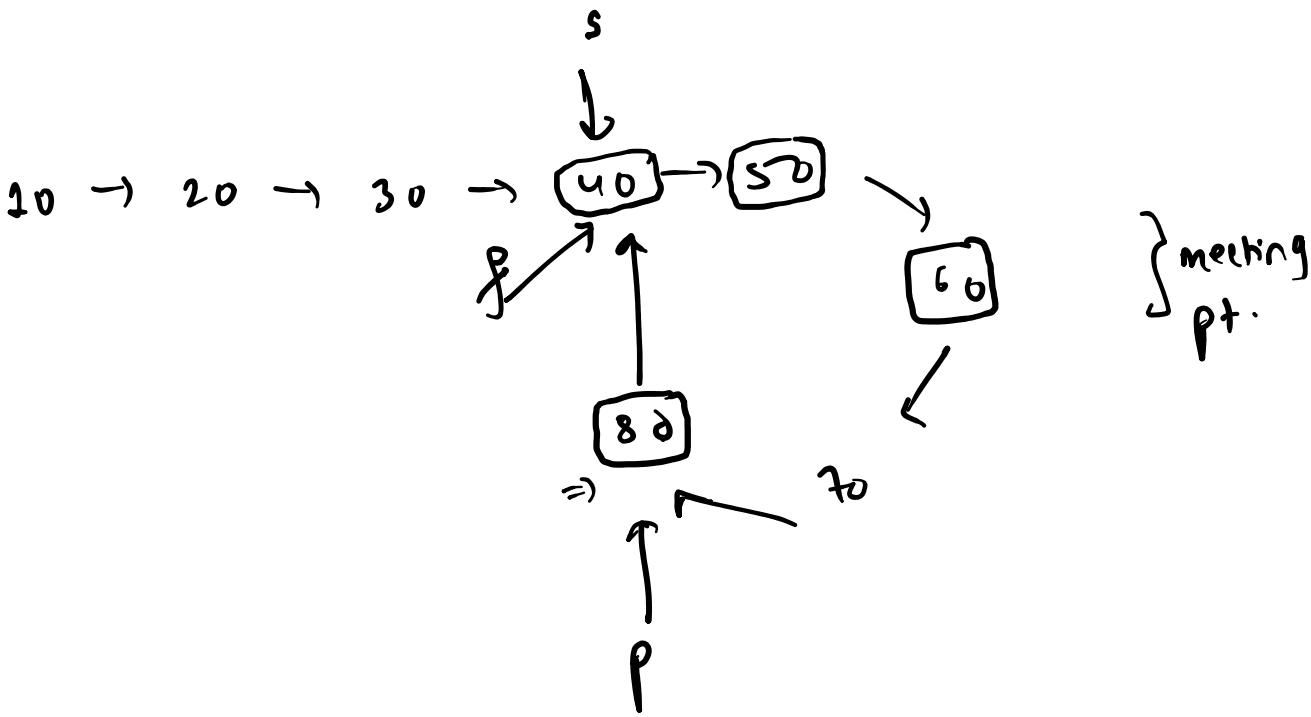


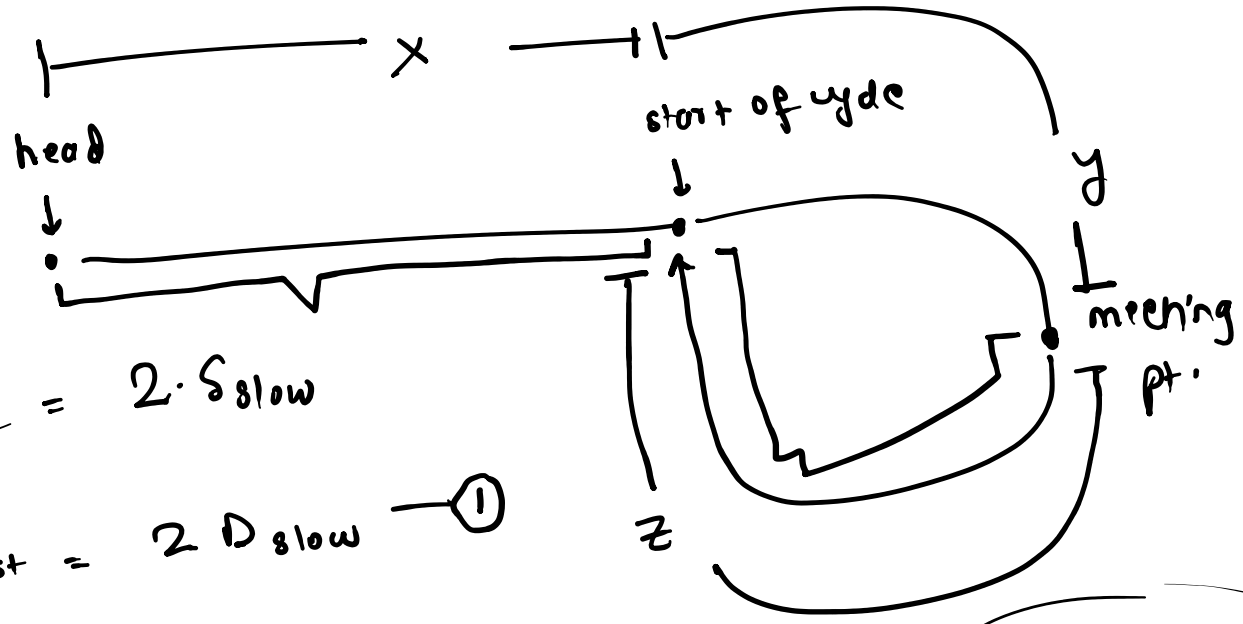



10 → 20 → 30 → 40 → 50
60 ← s
p
80
70


Remove Cycle from a Linked List
prev
curr
curr
curr
curr
curr
curr
100
200
300
400
500
600
10
200
20
300
30
400
40
500
50
600
60
300
100
prev
prev
prev
curr
prev
prev
prev


set 1
100 , 200 , 300
400 , 500 , 600


s
10 → 20 → 30 → 40 → 50
60
meeting pt.
80
70
p
p



$S_{fast} = 2 \cdot S_{slow}$

$\Rightarrow D_{fast} = 2 D_{slow}$ — (1)

$\therefore D_{fast} = x + y + z + y = x + 2y + z$

$\therefore D_{slow} = x + y$

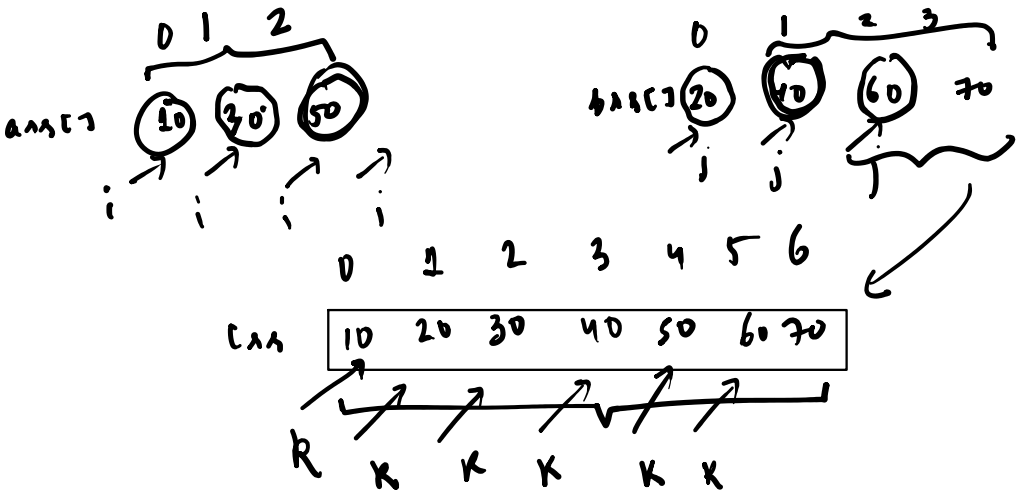
$x + 2y + z = 2(x + y)$

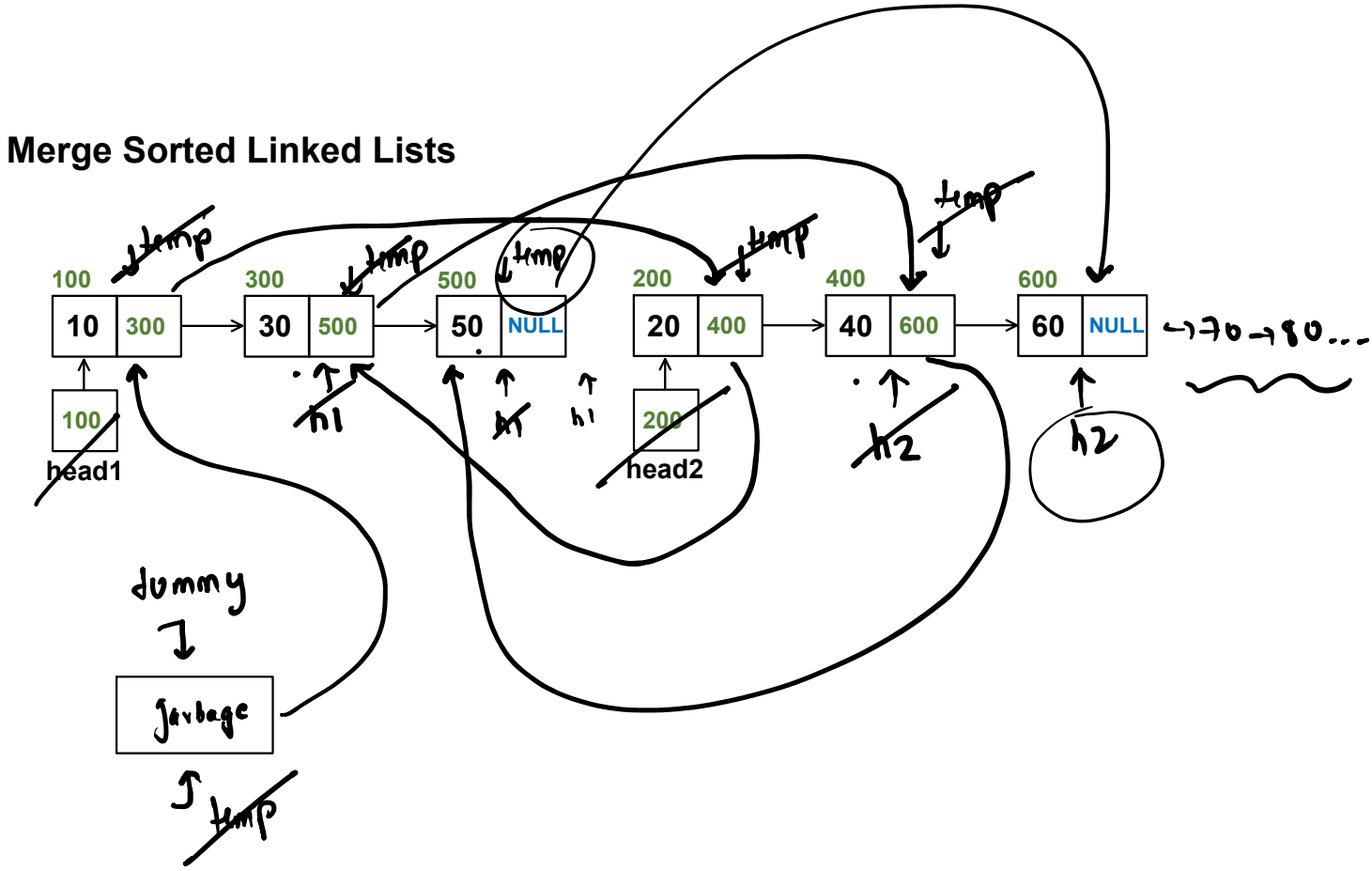
$x + 2y + z = 2x + 2y$

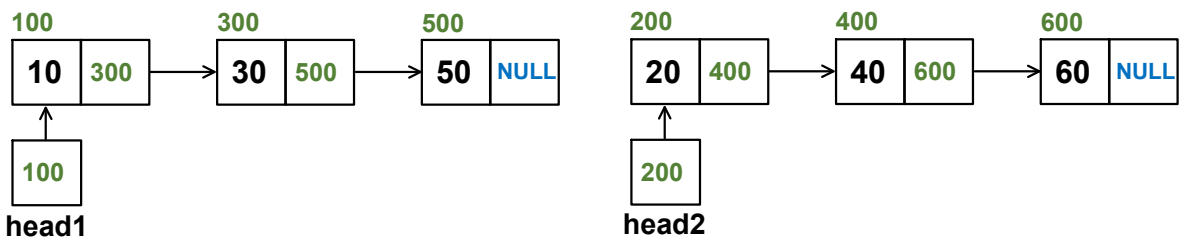
$\boxed{z = x}$

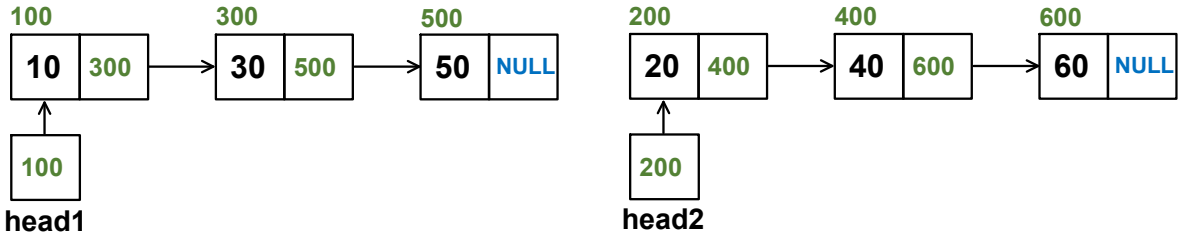
Merge Sorted Linked Lists

[HW] try to solve this question recursively

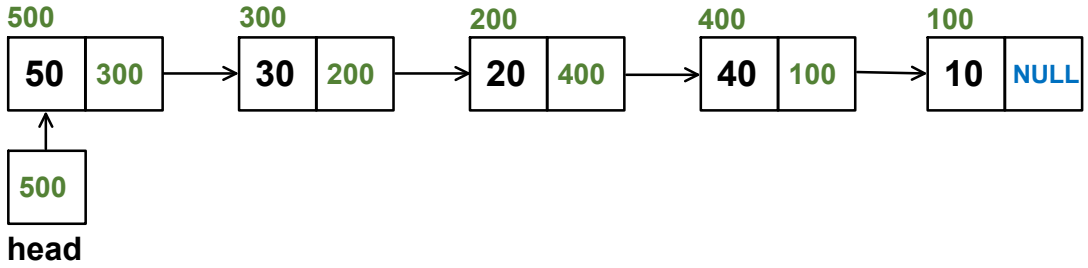


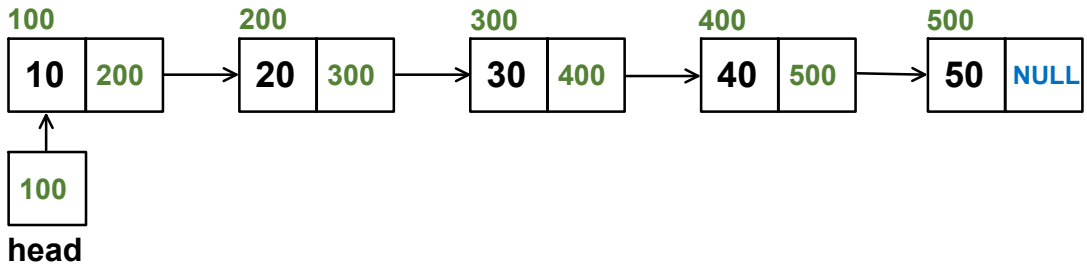





100
10
300
300
30
500
500
50
NULL
100
head1
200
20
400
400
40
600
600
60
NULL
200
head2

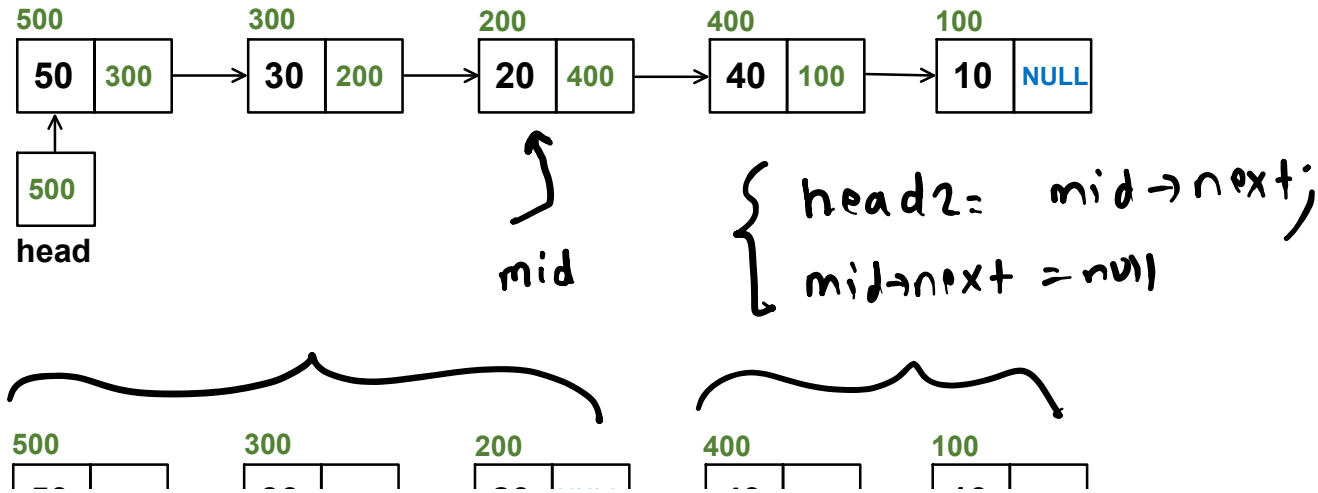
Merge-Sort a Linked List

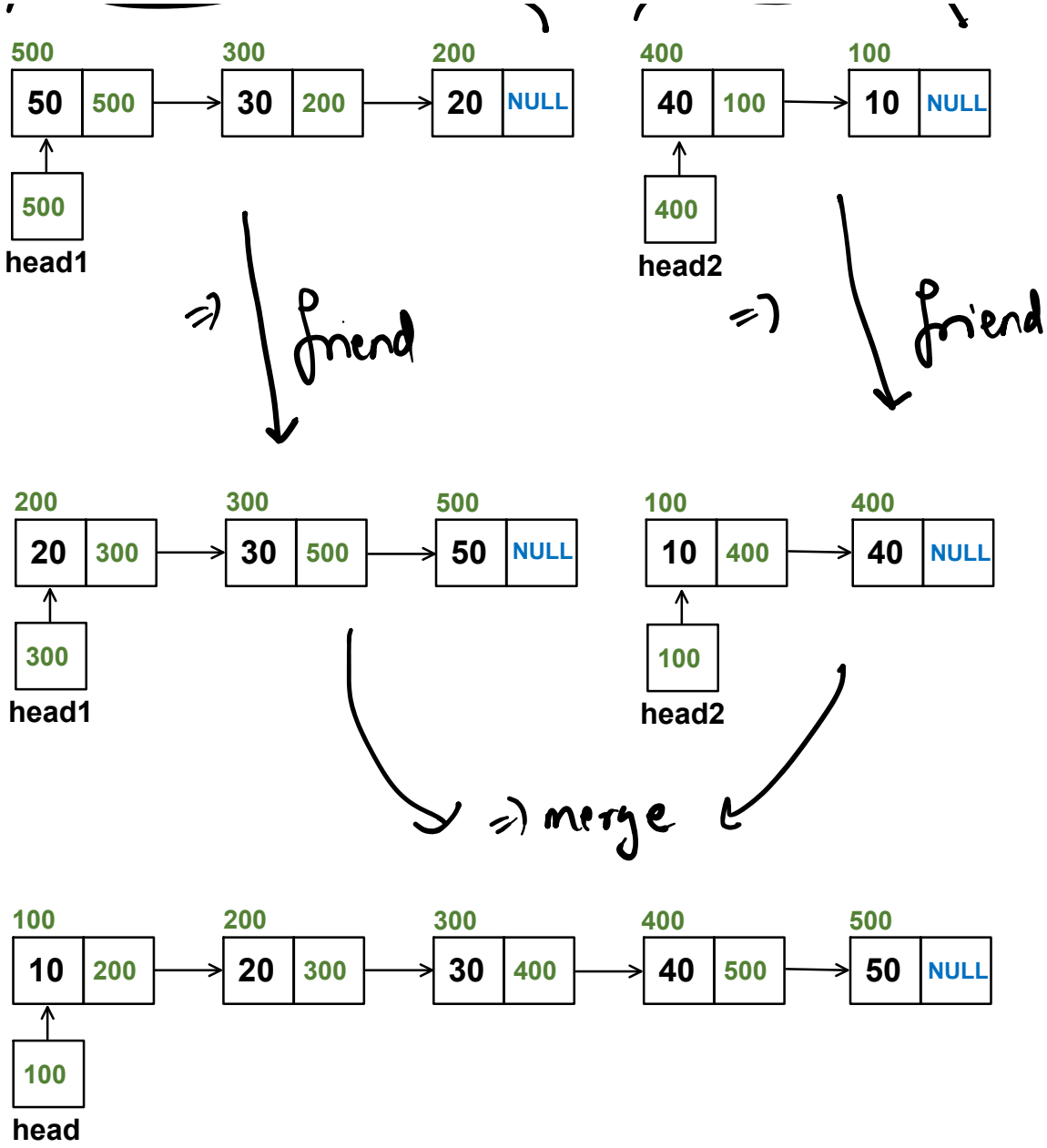




To **sort** a linked list using the **merge-sort** algorithm we follow three steps

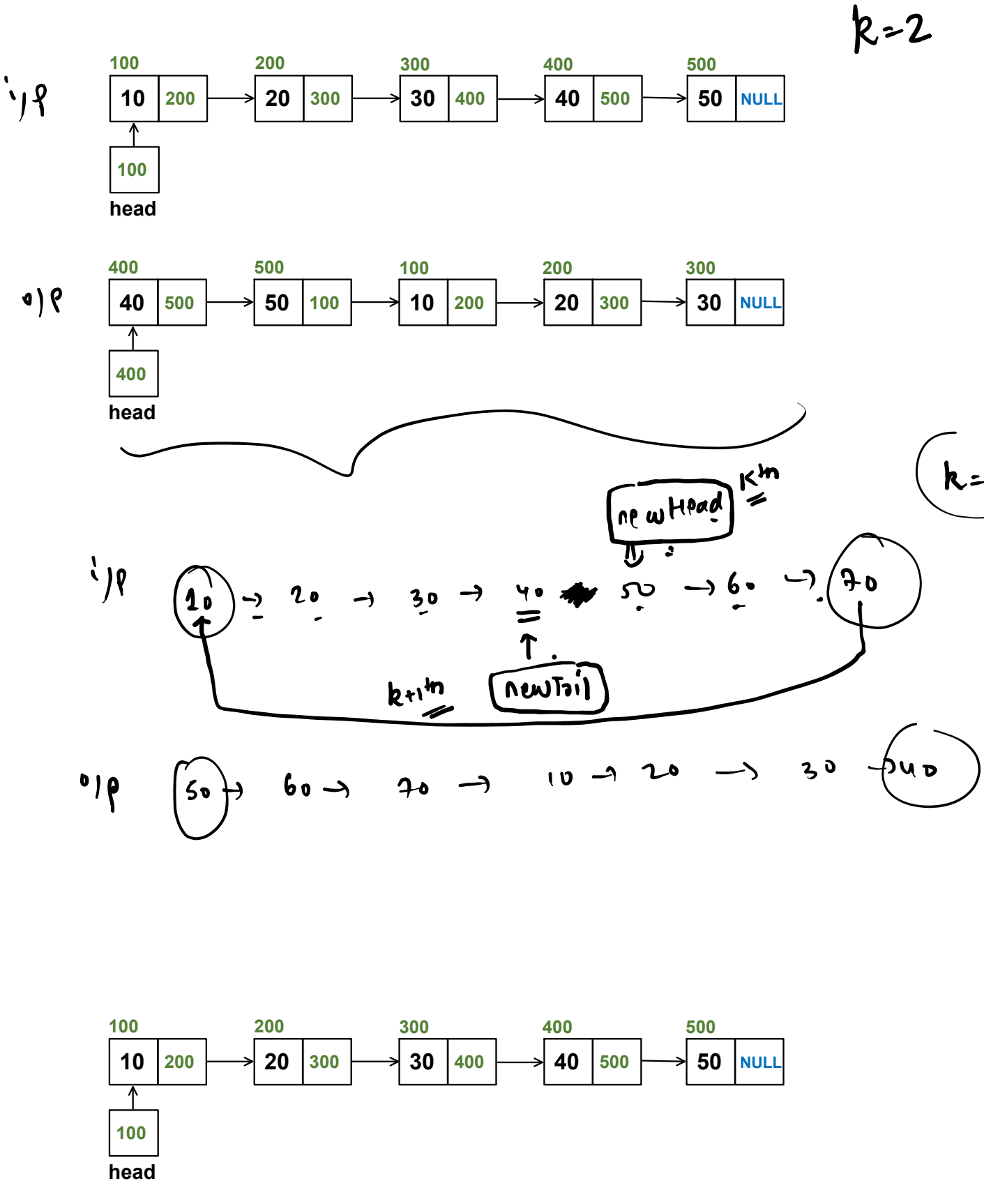
- **divide** the linked list into two sub-lists around the mid-point
- **recursively** sort the two sub-lists
- **merge** the two sorted sub-lists such that the merged list is sorted.




500
50
500
300
30
200
200
20
NULL
500
head1
400
40
100
100
10
NULL
400
head2
friend
friend
200
20
300
300
30
500
500
50
NULL
300
head1
100
10
400
400
40
NULL
100
head2
merge

100
10
200
200
20
300
300
30
400
400
40
500
500
50
NULL
100
head

K-Rotate a Linked List



[R][HW] Sum of Linked List

→ n_1 = 123

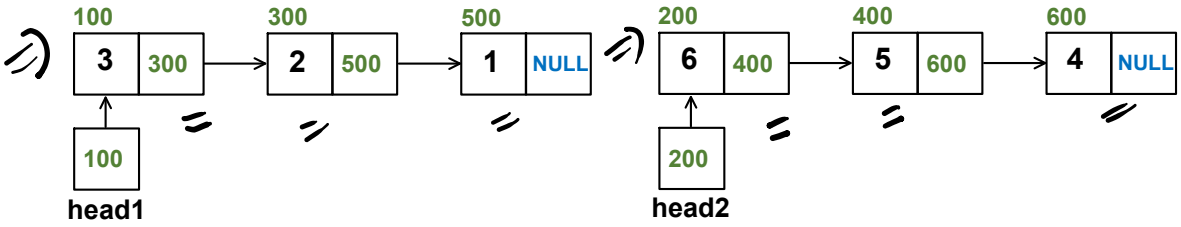
→ n_2 = 456

1 → 2 → 3

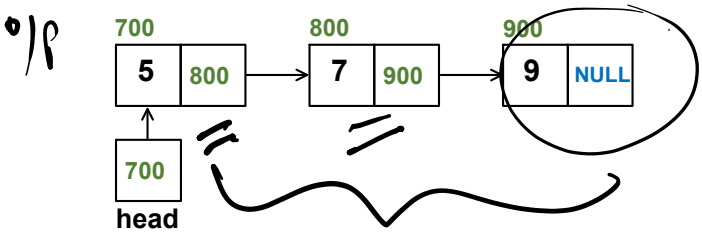
4 → 5 → 6

123.
+ 456
579

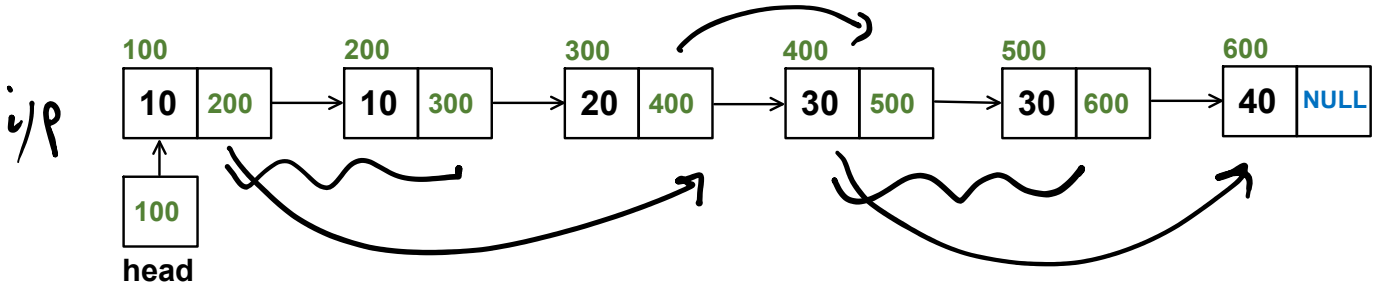
Sum of Linked List

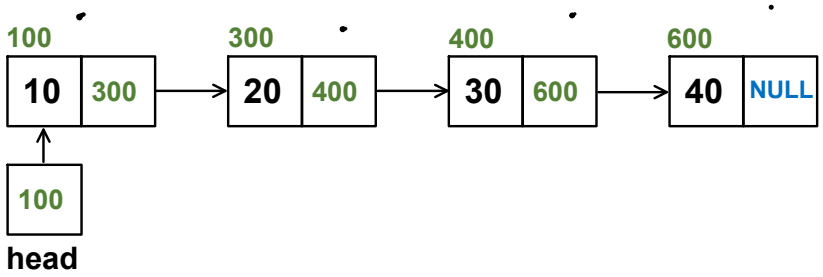


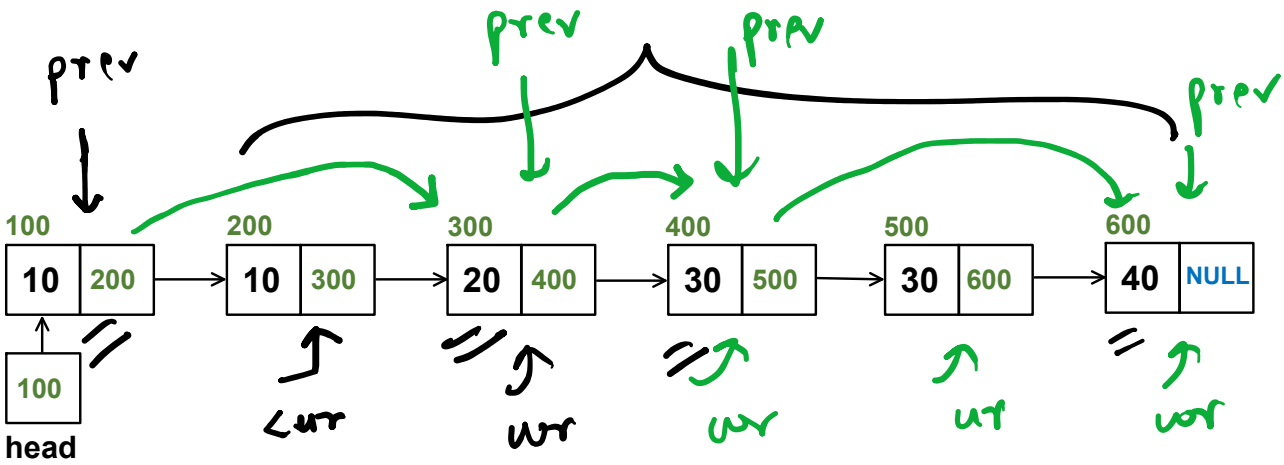
o/p



Remove Duplicates from Sorted Linked List I



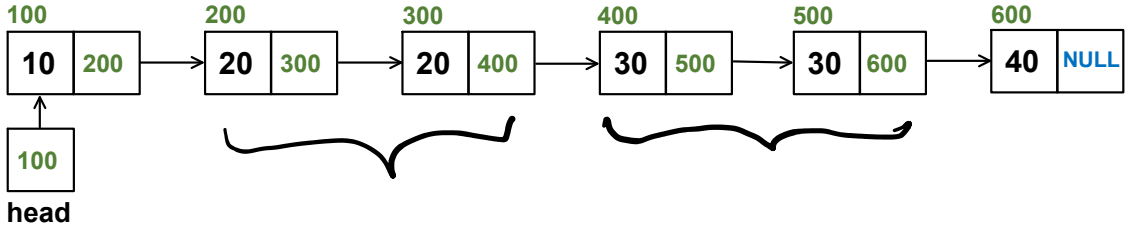




→ try to solve this recursively then try iterative

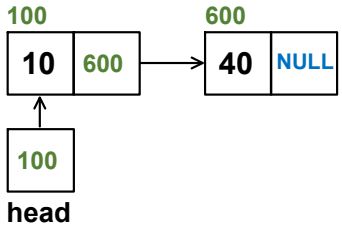
→ head may change

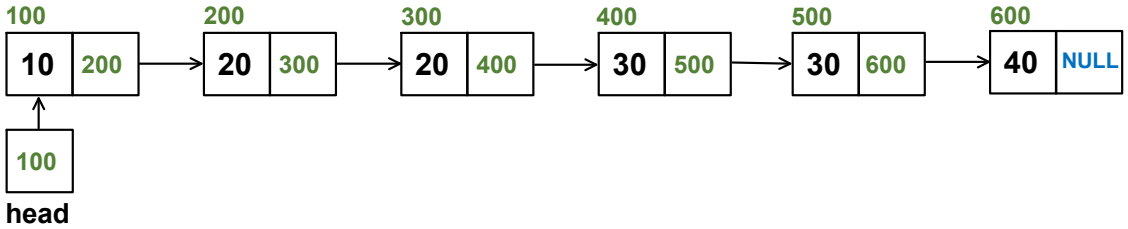
Remove Duplicates from Sorted Linked List II

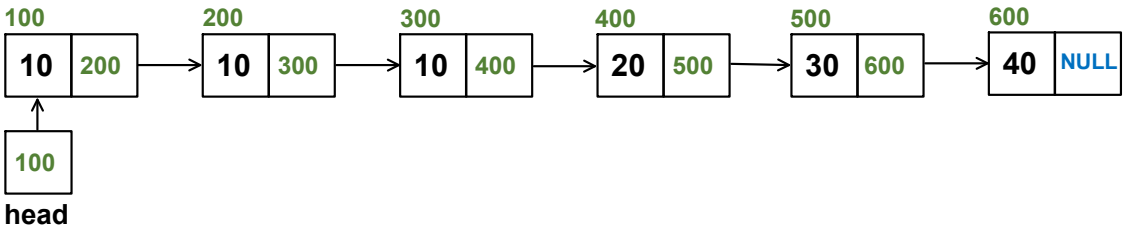


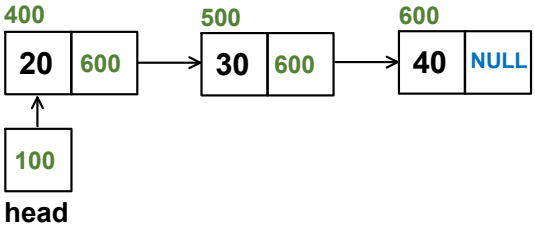
10 → 10 → 10 → 20 → 30 → 40

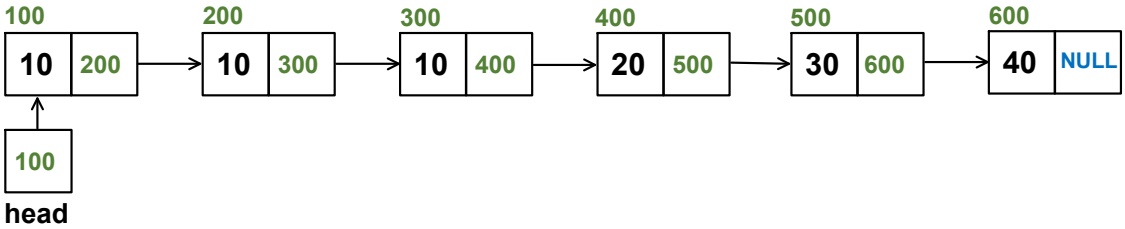
20 → 30 → 40








400
20
600
500
30
600
600
40
NULL
100
head


100
10
200
200
10
300
300
10
400
400
20
500
500
30
600
600
40
NULL
100
head

Bubble Sort a Linked List